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Original Research

Using a clinical narrative-aware pre-trained language model for predicting emergency department patient disposition and unscheduled return visits

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ABSTRACT

The increasing prevalence of overcrowding in Emergency Departments (EDs) threatens the effective delivery of urgent healthcare. Mitigation strategies include the deployment of monitoring systems capable of tracking and managing patient disposition to facilitate appropriate and timely care, which subsequently reduces patient revisits, optimizes resource allocation, and enhances patient outcomes. This study used ~ 250,000 emergency department visit records from Taipei Medical University-Shuang Ho Hospital to develop a natural language processing model using BlueBERT, a biomedical domain-specific pre-trained language model, to predict patient disposition status and unplanned readmissions. Data preprocessing and the integration of both structured and unstructured data were central to our approach. Compared to other models, BlueBERT outperformed due to its pre-training on a diverse range of medical literature, enabling it to better comprehend the specialized terminology, relationships, and context present in ED data. We found that translating Chinese-English clinical narratives into English and textualizing numerical data into categorical representations significantly improved the prediction of patient disposition (AUROC = 0.9014) and 72-hour unscheduled return visits (AUROC = 0.6475). The study concludes that the BlueBERT-based model demonstrated superior prediction capabilities, surpassing the performance of prior patient disposition predictive models, thus offering promising applications in the realm of ED clinical practice.

1. Introduction

Across the world, emergency departments (EDs) often suffer from overcrowding, which can delay critical medical care [1]. Since the prognosis of critical patients is relatively sensitive to timely interventions, early identification of patients in deteriorating condition is crucial [2,3]. Identifying patients who require hospitalization and the adequate medical care they need has always been a difficult task for emergency physicians. Clinical judgment, which is based on laboratory results along with physicians' past experience, might become misguided by individual factors and lead to increased morbidity and even mortality [4]. Therefore, developing a clinical decision support system regarding the prediction of the disposition of emergency patients can help to minimize human error, and also lead to a decrease in length of stay (LOS), improvement in the quality of medical care, and optimized allocation of the medical resources [5].

Current nurse-led triage systems can only represent the acuity of emergency patients, but not accurately identify the possible disposition [5]. Some studies have focused on improving patient flow in Eds [6], such as predicting admission demands by machine learning models, and have generated promising results [7–9]. However, these models were mainly based on structured data collected at ED, such as demographic data, vital signs, comorbidity and blood tests. That is, these prediction models were only able to reduce LOS when data were available soon after triage [10]. Besides these structured data (i.e., vital signs, demographic data), unstructured data are also collected at triage, such as the patients' chief complaints with related symptoms in a free-text form, and so are also available soon after triage. Compared with structured data, the chief complaints and present illness documented by triage nurses and clinical physicians can provide much more information and is included in the process of clinical decision-making. Recently, natural language processing (NLP) techniques have offered invaluable benefits

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in sifting through vast volumes of unstructured healthcare data, uncovering patterns and insights related to overcrowding causes. Predictive models using NLP can forecast overcrowding events and enable proactive interventions. Real-time monitoring with NLP allows healthcare facilities to manage patient flow, reduce congestion, and enhance patient care. We reviewed some studies that also dealt with predicting emergency department unscheduled revisit patients, which is shown in Table 1. Previous studies had established prediction models based on a combination of structured data and unstructured data with varied performances (AUROC from 0.61 to 0.76) and room for improvement [5,11,25,27]. Since the development of bidirectional encoder representations from transformers (BERT) were shown to improve classification accuracy compared to traditional methods [12,42], we introduce a BERT-based prediction model that effectively identifies possible hospitalized emergency patients.

Among patients who don't need hospitalization, around 2–5 % will return to ED within 72 h [13,14]. Around 19 %-30 % of these unscheduled revisit patients would be admitted to the hospital eventually while adverse events were found in one-third of this group [15,16]. Unnecessary tests and treatments with increased cost and length of stay at the ED were also found on the second ED visit [17]. Further, unscheduled revisits are often avoidable and potentially unnecessary [18]. According to previous studies, an unscheduled revisit is a multi-factored consequence, typically relating to one to five of the following factors: (1) poor quality patient care; (2) latent presentation of illness resulting in misdiagnosis; (3) poor communication between medical staff and patient at discharge; (4) patient uncertainty or fear of disease; and (5) insufficient management in the beginning [13,19–22]. Among these contributing factors, although the latent presentation of illness cannot be predicted, the remaining ones can be prevented through proper management. Medical staff will be able to discuss with patients to avoid premature discharge with proper education if they foresee the possibility of unscheduled revisits [23,24]. Moreover, if the physician is aware of the possibility of a patient's unscheduled revisit soon after triage, more medical instruction and management can be arranged at in the beginning. However, it is challenging to predict the possibility of early ED return, especially given its multi-factorial nature [25]. According to previous research, most informative contributing factors are collected at triage and can be extracted from electronic health records [26]. Therefore, our study's goal was to establish a model based on data collected at triage to make two predictions:

1) Whether patients will be hospitalized, and
2) Whether patients will return to ED within 72 h after discharge.

2. Materials and methods

2.1. Dataset and participants

This study used the electronic health records from the Emergency Department of Taipei Medical University-Shuang Ho Hospital, Ministry of Health and Welfare, from May 2017 to December 2019 for model development and consisted of a total of 245,721 emergency visits. Shuang-Ho Hospital is a teaching hospital located in New Taipei City with 95,000 ED visits annually.

The participants in the study were all adults who visited the emergency department and we excluded patients who sought medical treatment in response to administrative needs (e.g., issuing medical certificates, routine PCR examinations), patients with incomplete clinical records (e.g., no records of the patient's clinical history), pediatric and surgical patients. For the dataset of predicting patient disposition, we excluded patients who had suffered from Out-of-Hospital Cardiac Arrest (OHCA) and the patients with no record of disposition. For the dataset of predicting the patient's unscheduled return visits, we first screened out the medical records of the first medical visit within 72 h and excluded the patients who had cardiac arrest before the hospital and patients who were admitted to the hospital or discharged without the doctor's permission after the first visit. After excluding the above conditions, the final number of the dataset is 167,058, and 128,848 medical records for respectively predicting patient disposition and unscheduled return visits; these were used for the development of our model. The detailed data screening process is shown in Fig. 1. The patient information from the dataset included the patient's clinical history, such as chief complaints, past history, present illness and routine physiological measurements collected at triage. Demographic data such as gender, age, and mode of arrival (e.g., transfer from other hospitals, visit alone,

Table 1
Research of predicting emergency department unscheduled return visits.

Study	Location	Patient context	Sample size	URV rate	URV window	Approach	Variables	Model matrices
Cabezuelo, 2020 [27]	USA	ED general patients	290,000	4.32 %	3 days	Ensemble	Age, blood glucose, cardiac frequency, discharge medical time	AUC = 0.61 Acc = 0.957
Hong et al., 2019 [25]	USA	ED general patients	330,631	8.8 %	3 days	Gradient Boosting	Patient demographics, triage vitals, chief complaint (binary), etc.	AUC = 0.76 Sens = 0.16 Precision = 0.75
Davazdahemami, Peng, et al., 2022 [28]	USA	ED general patients	657,752	5.5 %	3 days	DNN	Patient demographics, vital sign, medical history, presenting illness, physical exam, etc.	AUC = 0.755 Acc = 0.680 Sens = 0.679
Davazdahemami et al., 2022 [29]	USA	ED COVID-19 patients	7,373	18 %	7 days	GA + DNN	Age, gender, race, medication, admission type, comorbidity	Acc = 0.874 Sens = 0.719 F_1-score = 0.704
Hong et al., 2019 [25]	USA	ED general patients	330,631	16 %	9 days	Gradient Boosting	Patient demographics, triage vitals, chief complaint (binary), etc.	AUC = 0.75 Sens = 0.23 Precision = 0.70
Hsu et al., 2022 [30]	Taiwan	ED patients with Abdominal Pain	449,594	5.85 %	3 days	Logistic Regression Random Forest Extreme Gradient Boosting	Patient demographics, vital sign, examinations	AUC = 0.74 Sens = 0.39 F_1-score = 0.25

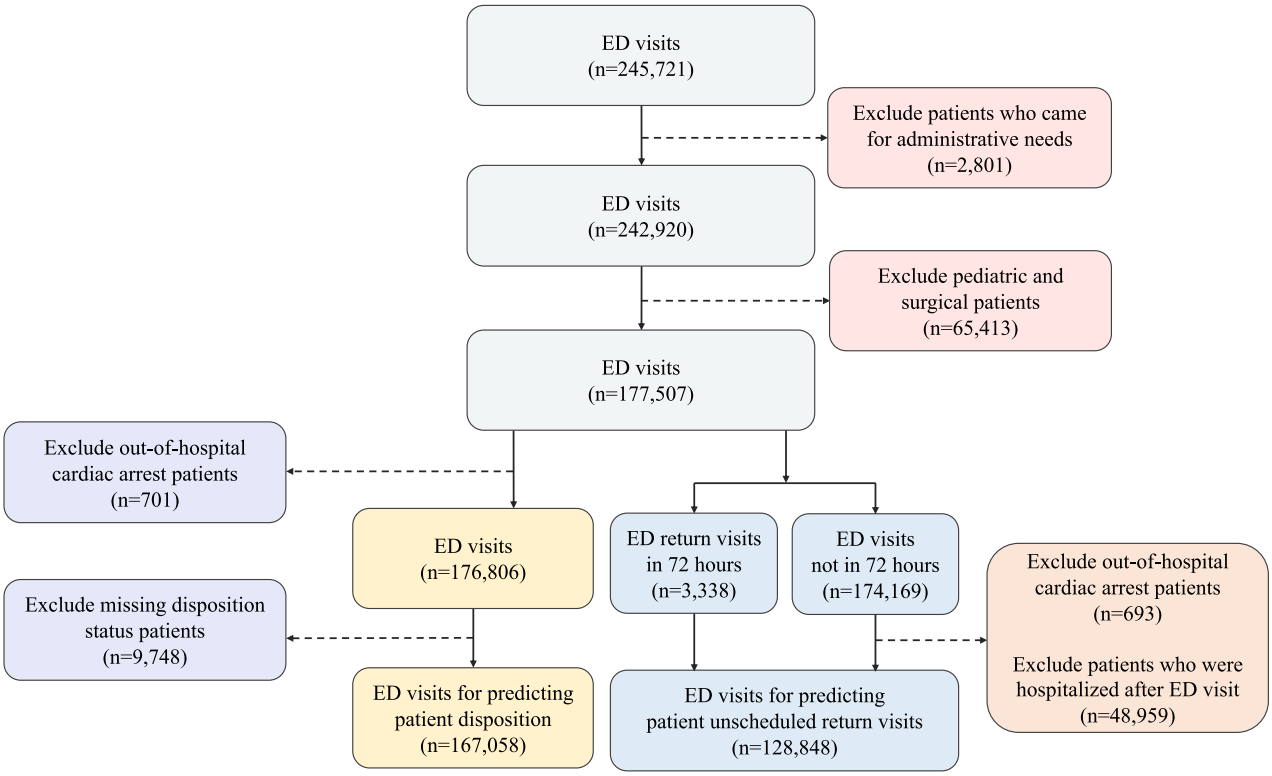

Fig. 1. Flow diagram of the dataset screening process.

or by ambulance) were collected. Also included were vital signs at triage including body temperature, pulse rate, respiratory rate, blood oxygen saturation, systolic blood pressure, and Glasgow Coma Scale (GCS) score. The clinical narratives included the chief complaints as documented by the triage nurse, and the chief complaints, present illness, and past history as documented by the emergency physician. To protect patient privacy, we only gathered information pertaining to gender and age, without collecting any individual identifiers; and other personal data of the patients were de-identified before conducting data analysis. This study was approved, and the informed consent process was waived, by the Human Investigation Committee at the authors' institution (IRB 2000022883).

Outcomes of the ED patient visit and predictor characteristics are shown in Table 2. We respectively used 167,058 and 128,848 visit records for predicting patient disposition and unscheduled return to the emergency department within 72 h of the visit. For the former, approximately 23.8 % of visits (39,721 visit records) resulted in hospitalization after the visit, and for the latter, about 2.2 % (2,887 visit records) of visits led to an unscheduled return to the emergency department within 72 h of the visit, which is similar to previous research in the USA (4.32 %) [27] and Taiwan (5.85 %) [28].

2.2. Data preprocessing

In the emergency department dataset, the *chief complaints* are a brief description of the patients' main symptoms and the reason for their visit. The present illness further describes relevant symptoms, such as pain location, duration, and related symptoms. The *past medical history* was recorded based on the patient's account. In Taiwan, the predominant practice among hospitals is to use Hospital Information Systems (HIS) for documentation purposes, which includes recorded notes, prescribed medications, and ordered examinations. Regarding the documentation process, once the patient's registration is complete, triage nurses undertake the measurement of vital signs and record the patient's chief complaints and the mode of arrival (e.g., walk-in, ambulance). During the subsequent physician consultations, symptoms and pertinent inquiries are documented and draw upon the clinician's expertise. All documented information is entered into the HIS system.

Effective mining and analysis of such clinical text data can provide valuable clinical diagnostic knowledge. compared to other countries, a distinguishing factor of Taiwan's healthcare documentation records is the linguistic context of medical practice. Although Mandarin and Taiwanese are the local languages, the foundation of medical education in Taiwan is heavily influenced by the United States, which means that medical knowledge and training are predominantly based on English textbooks and guidelines. As a consequence, medical documentation often features a mixture of Mandarin, English, and medical terminology.

However, based on our preliminary observations, we found significant issues of mixed Chinese and English text and misspellings in clinical narratives. In addition, the inconsistent data recording methods, such as irregular spacing or punctuation in text records, the use of different units for physiological measurements, and the use of different abbreviations for the clinical terms, further complicated data processing. Variations in physicians' individual writing habits also result in personalized narrative styles or informal colloquial writing, as well as a variety of abbreviations and symbol usages, such as "*◎Chief Complaint:, 沒有腎臟病, NKDA, occupation: nil, cluster//travel hx: nil, contact hx: 昨天看護疑似有發燒, abd pain? no 吐拉, R-N (−), sorethroat (?), PI: Cough(+) chronic 沒明顯變嚴重, fever noted today*".

The complexity of the clinical text data mentioned above can affect the learning effectiveness of the model. Therefore, to ensure the quality of the data, we carefully preprocessed it. We used the Python library to detect common spelling and grammatical errors, which were then verified by clinical doctors. In this way, we corrected spelling errors and abbreviations, such as correcting "L/W" to "laceration" and "Parkenson" to "Parkinson's Disease." In addition, certain punctuation marks may be used in medical terminology, where some clinicians tend to use "+" or "-" to indicate the presence or absence of symptoms. For instance, "dizzy (+)" indicates dizziness in a patient. As such, we converted common symbols into text descriptions, such as transforming "fever(−)" into "no

Table 2
Dataset descriptive statistics.

Variable	Disposition (n = 167,058)		URV (n = 128,848)	
	MBD n = 127,337 (76.2 %)	Hospitalized n = 39,721 (23.8 %)	Yes n = 2,887 (2.2 %)	No n = 125,961 (97.8 %)
Age				
Young adult (18 ~ 40 years old)	51,950 (40.80 %)	6,166 (15.52 %)	1,099 (38.07 %)	51,349 (40.77 %)
Mid-age adult (41 ~ 64 years old)	45,815 (35.98 %)	12,535 (31.56 %)	957 (33.15 %)	45,406 (36.05 %)
Elderly (>65 years old)	29,572 (23.22 %)	21,020 (52.92 %)	831 (28.78 %)	29,206 (23.19 %)
Gender				
Male	57,570 (45.21 %)	21,048 (52.99 %)	1,361 (47.14 %)	56,946 (45.21 %)
Female	69,767 (54.79 %)	18,673 (47.01 %)	1,526 (52.86 %)	69,015 (54.79 %)
Mode of Arrival				
Transfer from external hospital	6,460 (5.07 %)	2,996 (7.54 %)	148 (5.13 %)	6,396 (5.08 %)
From 119 Ambulance	9,778 (7.68 %)	9,383 (23.62 %)	276 (9.56 %)	9,686 (7.69 %)
Visit alone	111,099 (87.25 %)	27,342 (68.84 %)	2,463 (85.31 %)	109,879 (87.23 %)
Temperature				
Low (<35.0℃)	104 (0.08 %)	172 (0.43 %)	3 (0.10 %)	107 (0.10 %)
Normal (35 ~ 38℃)	117,464 (92.25 %)	32,862 (82.73 %)	2,588 (89.60 %)	116,149 (92.20 %)
High (>38.0℃)	9,769 (7.67 %)	6,687 (16.83 %)	296 (10.30 %)	9,705 (7.70 %)
Systolic Blood Pressure				
Low (<100 mm/Hg)	3,356 (2.64 %)	2,822 (7.10 %)	89 (3.08 %)	3,353 (2.66 %)
Normal (100 ~ 180 mm/Hg)	115,452 (90.67 %)	33,305 (83.85 %)	2,569 (88.99 %)	114,213 (90.67 %)
High (>180 mm/Hg)	8,529 (6.70 %)	3,594 (9.05 %)	229 (7.93 %)	8,395 (6.66 %)
Pulse Rate				
Low (<60 times/minutes)	3,377 (8.04 %)	1,516 (3.82 %)	70 (2.42 %)	3,343 (2.65 %)
Normal (60 ~ 100 times/minutes)	94,815 (22.57 %)	24,052 (60.55 %)	2,033 (70.42 %)	93,735 (74.42 %)
High (>100 times/minutes)	29,145 (69.39 %)	14,153 (35.63 %)	784 (27.16 %)	28,883 (22.93 %)
Respiratory Rate				
Low (<12 times/minutes)	3 (0.00 %)	38 (0.10 %)	0 (0.00 %)	8 (0.01 %)
Normal (12 ~ 20 times/minutes)	122,067 (95.86 %)	32,876 (82.77 %)	2,738 (94.84 %)	120,702 (95.82 %)
High (>20 times/minutes)	5,267 (4.14 %)	6,807 (17.14 %)	149 (5.16 %)	5,251 (4.17 %)
Oxygen Saturation				
Low (<90 %)	112 (0.09 %)	1,654 (4.16 %)	8 (0.28 %)	127 (0.10 %)
Normal (90 ~ 100 %)	127,225 (99.91 %)	38,067 (95.84 %)	2,879 (99.72 %)	125,834 (99.90 %)
GCS Total Score				
Severe (score = 3 ~ 8)	671 (0.53 %)	3,083 (7.76 %)	32 (1.11 %)	694 (0.55 %)
Mild (score = 8 ~ 14)	2,980 (2.34 %)	5,105 (12.85 %)	98 (3.39 %)	2,963 (2.35 %)
Normal (score = 15)	123,686 (97.13 %)	31,533 (79.39 %)	2,757 (95.50 %)	122,304 (97.10 %)

fever", "dizzy (+/-)" into "mild dizzy", and "diarrhea(+)" into "diarrhea".

2.3. Model architecture

The architecture of our model is shown in Fig. 2. Since clinical narratives can offer a valuable source of medication information for physicians in making accurate diagnoses or treatment options, this study focused on processing text and extracting the most critical information from clinical data to maximize the learning performance of the model. However, the presence of incomplete and ungrammatical sentences was frequently observed, and the inclusion of Chinese characters was prevalent, considering that Chinese serves as the official working language in Taiwan, with Mandarin being extensively used. This presented an analytical obstacle since most conventional NLP algorithms are primarily tailored for monolingual data processing. The incorporation of multilingual texts can significantly impair the prediction performance of the model, as it is constrained by the grammar rules of a single language [29]. Drawing from prior research, we found that unifying the language of textual data enhances the model's comprehension and predictive capabilities [41]. This is why we translated Chinese words into English using the Google Cloud Platform to achieve text consistency for the problems involving combinations of English and Chinese, such as "No 拉肚子" to "no diarrhea". We chose the Google Cloud Platform Translation API to translate Chinese characters into English words, due to its high accuracy, widespread industry adoption, and advanced machine learning that offers contextually nuanced translations [43]. This was important for maintaining the integrity of medical texts. Through continuous updates and a vast multilingual corpus, including healthcare-specific terms, we ensured our translations were up-to-date and semantically accurate. This significantly reduces the risk of discrepancies in our analysis.

Vital signs also play an important role in assessing a patient's overall health status in emergency medicine. In this research, the principles of the conversion of each physiological measurement value are based on universally accepted clinical definitions. Taking heart rate as an example, the normal range for an individual's heart rate is typically between 60 and 100 beats per minute. Heart rates exceeding 100 beats per minute are classified as tachycardia, while rates below 60 beats per minute are classified as bradycardia. Such classification criteria are widely accepted in clinical medicine, which are applied in our research. Another example pertains to consciousness assessment, where a full Glasgow Coma Scale (GCS) score is 15 points, with scores of 3–8 indicating severe impairment, and scores of 9–14 denoting mild impairment. This classification system is extensively utilized in the evaluation of patients with head trauma and serves as a basis for clinical decision-making. In our research, this classification was also applied to convert the consciousness measurement into a narrative description. Overall, the principles of converting the physiological measurements are based on the clinical definition of abnormal values.

Insofar as vital sign data consisting of numerical values exhibit significant scale differences, this study standardized the data by converting numerical data into categorical data and comparing the effectiveness of the two types of data, i.e., Numeric Data (ND) and Categorical Numeric Data (CND). We categorized patients' age into 3 groups: young adults (18–40 years), middle-aged adults (41–64 years), and the elderly (≥65 years). One-hot encoding was used for gender and arrival type. Based on the recommendations of clinical physicians, we categorized vital sign features into "low, normal, high" or "abnormal and normal" levels. The detailed categorization method is shown in Table 2. In order to effectively integrate numerical vital signs with language models, we further converted numerical data into text descriptions using a linguistic template. For instance, "The patient is *elderly*, *female*, arrival type was by *came alone*. The body temperature is *high*, systolic blood pressure is *normal*, pulse rate is *high*, repository rate is *low*, pulse oximetry is *normal*, the total coma score is *normal*." For this type of data, we named it Textualized Categorical Numeric Data (TCND). The detailed explanations and examples for each setting of input data are shown in Tables 3, and 4 shows the detailed summary statistics (mean, std dev, q1, q3) for the input features, including character length, word length, and token length (as processed by BERT tokenizer).

The purpose of this study is to effectively predict emergency patient disposition and unscheduled return visits from clinical narratives. To

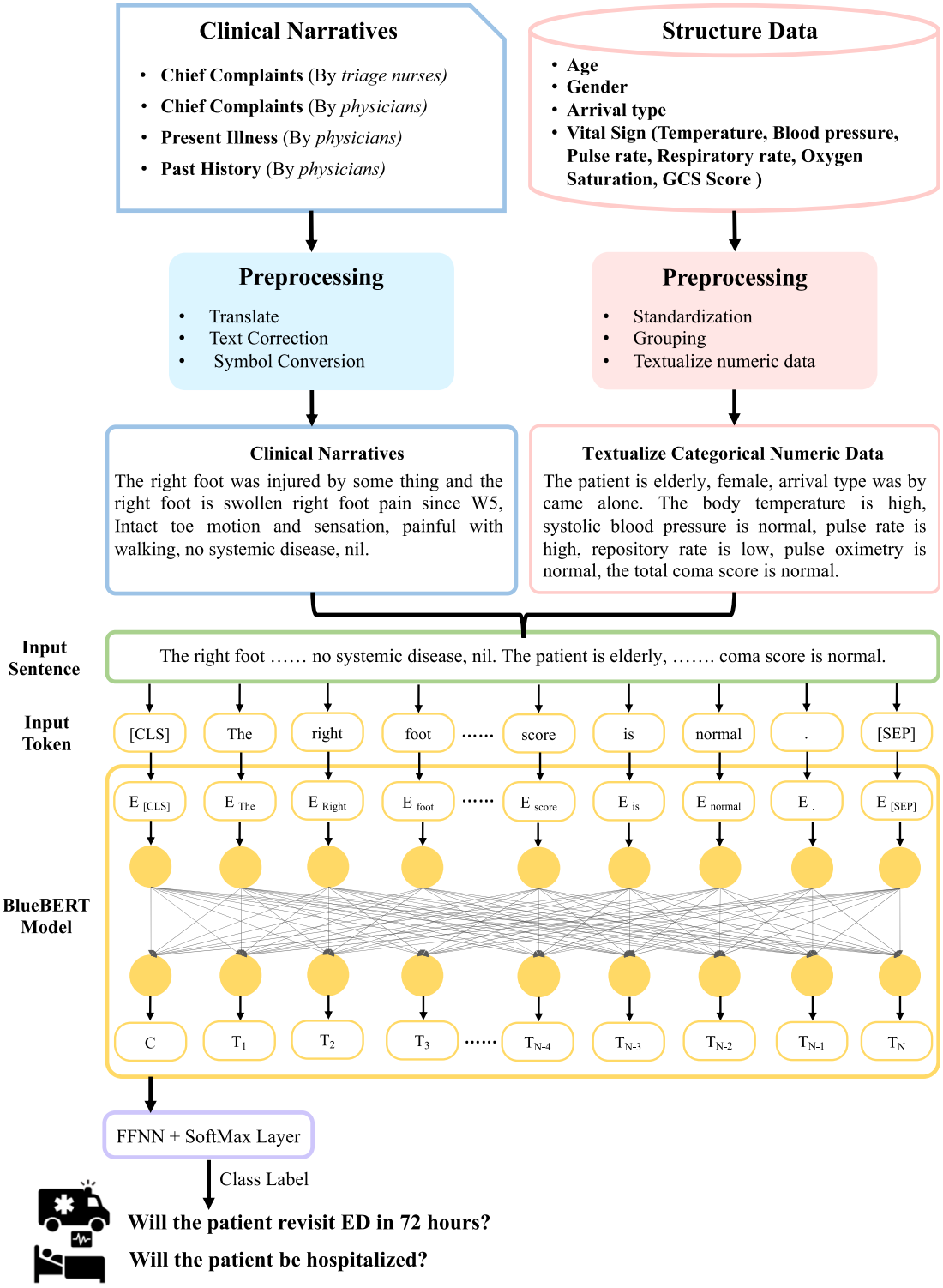

Fig. 2. Architecture of the clinical narrative-aware pre-trained language model for ED patient disposition and return visit prediction.

accomplish this, we used the BlueBERT [30] (Biomedical Language Understanding Evaluation Bidirectional Encoder Representations from Transformers) model, which is a widely adopted pre-trained language model specifically designed for biomedical applications. In the initial experiment, we also utilized BioClinical BERT, which is also a pre-trained language model fine-tuned on clinical and biomedical text data, but BlueBERT performed better than BioClinical BERT so we finally used BlueBERT for subsequent experimental comparisons. Initially, we employed the WordPiece toolkit [31] to tokenize the text and create sequences of tokens. The token embeddings were then generated using the pre-trained BlueBERT model, which effectively represented the words within the clinical narratives. We also incorporated positional embeddings to capture the sequential information of the tokens [32]. Following the established conventional use of the pre-trained BERT-based model, the first token in the input text sequence was assigned as a distinct token [CLS], and the representation of the text

Table 3
Examples of input data.

Data type	Input examples
Baseline	病患來診為右腳被東西用傷現右腳腫 Right foot pain since W5. Intact toe motion and sensation, painful with walking. systemic disease(−), nil(+).
TT	The right foot was injured by something and the right foot is swollen. Right foot pain since W5. Intact toe motion and sensation, painful with walking. No systemic disease, with nil.
TT + ND	The right foot was injured by something and the right foot is swollen. Right foot pain since W5. Intact toe motion and sensation, painful with walking. No systemic disease, with nil. + [80, 1, 0, 0, 1, 0, 38.5, 120, 105, 10, 98, 15]
TT + CND	The right foot was injured by something and the right foot is swollen. Right foot pain since W5. Intact toe motion and sensation, painful with walking. No systemic disease, with nil. + [80, 1, 0, 0, 1, 0, 2, 1, 2, 0, 1, 2]
TT + TCND	The right foot was injured by something and the right foot is swollen. Right foot pain since W5. Intact toe motion and sensation, painful with walking. No systemic disease, with nil. + The patient is *elderly*, *female*, arrival type was by *came alone*. The body temperature is *high*, systolic blood pressure is *normal*, pulse rate is *high*, repository rate is *low*, pulse oximetry is *normal*, the total coma score is *normal*.“

Baseline = Non-translated Text, TT = Translated Text, ND = Numeric Data, CND = Categorical Numeric Data, TCND = Textualize Categorical Numeric Data.

Table 4
Summary statistics of input features.

Data	Data type	Text type	Mean	STD	Q1	Q3
Unscheduled Return Visits	TranslateText	Character	289.90	135.42	198	365
		Word	44.70	21.52	30	56
		Token	79.96	37.49	54	101
	Numeric Data to Text	Character	239.50	4.81	237	241
		Word	39.46	0.74	39	40
		Token	53.19	0.94	53	54
	TranslateText + Numeric Data to Text	Character	540.58	142.75	443	619
		Word	87.58	24.95	71	101
		Token	138.32	42.11	109	161
Disposition	TranslateText	Character	302.83	144.47	205	380
		Word	46.73	22.97	31	59
		Token	83.34	39.82	56	105
	Numeric Data to Text	Character	239.09	5.18	236	241
		Word	39.42	0.77	39	40
		Token	53.14	1.01	52	54
	TranslateText + Numeric Data to Text	Character	553.13	151.41	450	643
		Word	89.54	26.27	72	103
		Token	141.71	44.31	111	165

was learned through special tasks during the training process. When performing specific tasks, the [CLS]-tagged output vector was used as a summary representation of the entire input text sequence and then fed into a classifier or another task-specific layer for specific classification or prediction [12,42]. After this, we proceeded to unfreeze all parameters and conducted additional training epochs. This resulted in the generation of a 768-dimensional vector for text representation, comprising token embeddings, positional embeddings, and keyword embeddings. The multi-head attention mechanism [32] was employed to facilitate the learning process by identifying the specific components of the representation that required greater focus.

Using the multi-head attention mechanism, vectors were mapped to yield a weighted sum of the values, with weights determined by a compatibility function between the query vector and the key vector. For our specific study, we denoted the matrices of query, key, and value as Q, K, and V, respectively, where each matrix has dimensions of $\mathbb{R}^{d_a}$. The attention operation described in Equation (1) was employed to derive the output matrix [32]. In this research, we used multiple Transformer layers with multi-head attention instead of a single attention function, which allowed our model to simultaneously attend to information from different subspaces of representation at various positions using the multi-head attention mechanism [32]. Further, each attention head handles a different objective of the input sentence, i.e., patient disposition and unscheduled return visits. To leverage the benefits of the pre-trained BERT-based model, we set the number of Transformer layers to 12, hidden size to 768, and the number of attention heads per layer to 12. In the next layer, fully-connected layers are gradually applied to reduce the output dimension to two for prediction.

$$Attention(Q, K, V) = softmax\left(\frac{QK^T}{\sqrt{d_a}}\right)V \tag{1}$$

As shown in Table 2, the dataset is imbalanced, which is a common issue in clinical data [33] that not only severely affects the learning effectiveness of the model but also reduces the accuracy of the model's predictions. Therefore, this study used the random under-sampling method (RUS) to adjust the label proportion of the training data, reducing the excessive imbalance in the two classes of labels of training data that would affect model learning. Through our experiments, we found that adjusting the ratio of the two labels to 1:3 could make the model achieve the best-predicting performance. Furthermore, the proposed model was implemented using PyTorch,[2] a deep learning library in Python. We followed the original optimization and hyper-parameter settings for fine-tuning, which involved training for 10 epochs and using the *AdamW* optimizer [34]. The learning rate was set to 2e-5, while the β_2 value was set to 0.98 to enhance training stability. The maximum sequence length allowed was 512 tokens, with padding or truncation applied at the end of the sequence as needed. Our model was executed on a single Nvidia GeForce RTX 3090 Ti GPU.

2.4. Comparative analysis models and evaluation metrics

In this research, we analyzed five machine learning models and two large pre-trained language models to comprehensively evaluate the prediction model. The machine learning models included Complement Naïve Bayes (CNB), K-Nearest Neighbors (KNN), Random Forest (RF), XGBoost (XGB), and LightGBM (LGBM). For the large pre-trained language models, we implemented BioClinical Bidirectional Encoder Representations from Transformers (BioClinical BERT) [35] and BlueBERT [30], which are both widely used in the biomedical informatics and clinical domain. Note that our study was based on untranslated clinical texts and excluded categorical variables from our machine learning comparisons. This approach ensures a focused evaluation of text-based inputs in our models and maintains consistency with translated clinical data, thereby omitting categorical variables from our ML baselines. To evaluate the performance of the seven models and determine which model is most effective for the task at hand, we included the area under the receiver operating curve (AUROC), the area under the precision-recall curve (AUPRC), sensitivity, specificity, positive predictive value (PPV), and negative predictive value (NPV) in our experiments. In addition to establishing and verifying a theoretical model, we also hope the product of this research can be applied in clinical practice. Therefore, in terms of model verification, a 10-fold cross-validation approach was adopted to evaluate the model performance.

3. Results

Our experiment investigated the effectiveness of different processing methods applied to clinical data, as shown in Table 5. Firstly, we compared the performance of translating mixed English and Chinese clinical texts. In terms of AUROC and AUPRC scores, we observed an improvement in model prediction performance when the text was

[2] https://pytorch.org/.

Table 5
Evaluating the incremental contribution of the proposed method.

Experiment	Disposition						Unscheduled Return Visits					
	AUROC	AUPRC	Sensitivity	Specificity	PPV	NPV	AUROC	AUPRC	Sensitivity	Specificity	PPV	NPV
BlueBERT (Baseline)	0.8798 (±0.0038)	0.7349 (±0.0082)	0.6238 (±0.0115)	0.9208 (±0.0025)	0.7106 (±0.0063)	0.8870 (±0.0030)	0.6334 (±0.0157)	0.0510 (±0.0056)	0.2172 (±0.0378)	0.9085 (±0.0192)	0.0522 (±0.0057)	0.9806 (±0.0006)
+ TT	0.8891 (±0.0031)	0.7513 (±0.0092)	0.7948 (±0.0087)	0.8561 (±0.0036)	**0.7948** (±0.0098)	0.9201 (±0.0024)	0.6455 (±0.0171)	0.0561 (±0.0046)	0.2092 (±0.0406)	**0.9192** (±0.0194)	0.0567 (±0.0040)	0.9807 (±0.0006)
+ TT + ND	0.7922 (±0.0046)	0.7303 (±0.0059)	0. 7987 (±0.0088)	0. 8574 (±0.0028)	0.6679 (±0.0078)	0.9164 (±0.0024)	0.5696 (±0.0128)	**0.1554** (±0.0187)	0.2421 (±0.0368)	0.8971 (±0.0196)	0.0518 (±0.0063)	0.9810 (±0.0006)
+ TT + CND	0.7873 (±0.0022)	0.7283 (±0.0071)	0.7966 (±0.0119)	0.8574 (±0.0039)	0.6536 (±0.0084)	**0.9210** (±0.0031)	0.5668 (±0.0104)	0.1488 (±0.0544)	0.2262 (±0.0574)	0.9074 (±0.2120)	0.0540 (±0.0126)	0.9808 (±0.2341)
+ TCND + TT	**0.9037** (±0.0031)	**0.7797** (±0.0067)	0.6650 (±0.0102)	**0.9273** (±0.0027)	0.7405 (±0.0067)	0.8987 (±0.0027)	0.6413 (±0.0122)	0.0514 (±0.0058)	0.1867 (±0.0488)	0.9336 (±0.0204)	**0.0623** (±0.0083)	0.9804 (±0.0007)
+ TT + TCND (Our method)	0.9014 (±0.2155)	0.7739 (±0.1855)	**0.8037** (±0.1630)	0.8608 (±0.2290)	0.6760 (±0.1783)	0.9184 (±0.2233)	**0.6475** (±0.0202)	0.0539 (±0.0056)	**0.2435** (±0.0251)	0.8991 (±0.0132)	0.0532 (±0.0081)	**0.9811** (±0.0006)

Baseline = Non-translated Text, TT = Translated Text, ND = Numeric Data, CND = Categorical Numeric Data, TCND = Textualize Categorical Numeric Data.

translated into English, with AUROC score of predicting disposition from 0.8798 to 0.8891 and predicting URV from 0.6334 to 0.6455. The improved performance of the model after translating text data and unifying the language has been shown in other experiments [41]. Regarding numerical comparisons, we evaluated the numerical data by considering it as continuous or categorical, or by transforming categorical data into text. The results demonstrate that converting categorical numerical data into text can boost the performance in predicting patient disposition and unscheduled return visits. In this study, we also compared two methodologies for integrating numeric data into the predictive model: one approach involved pre-pending the clinical narrative with numeric data (+TCND + TT), and the other appended it post-narrative (+TT + TCND). In tests comparing both methodologies, the performance of AUROC and AUPRC was comparable. Notably, the + TT + TCND method exhibited a significant enhancement in sensitivity, indicating an improved capability in the correct identification of true positive samples. This improvement is particularly salient for clinical settings where sensitivity is prioritized over specificity, especially when the reduction of false negatives is paramount. Further analysis revealed that the sequencing of vital sign data integration impacts the model's learning of feature representations, thus affecting sensitivity to different sample types. In light of these findings, we advocate for the + TT + TCND approach as the preferred strategy. This choice is not merely due to its superiority in sensitivity but also because of its superior performance in predicting unscheduled return visits, a key factor in preventing adverse clinical outcomes and enhancing patient care quality.

For the comprehensive evaluation of the proposed method, we further compared it with seven well-known methods, including machine learning, deep learning, and pre-trained language models. The comparison results are presented in Table 6 and Fig. 3. Two large pre-trained language models, BioClinicalBERT and BlueBERT demonstrate significantly superior performance to the other compared machine learning-based methods. This highlights the capability of pre-trained BERT models to learn the most crucial information from text. Furthermore, since these two BERT models are pre-trained on clinically relevant corpora, they are more capable of capturing the most crucial clinical insights within the text. It is also worth mentioning that among the two language models, the performance of BlueBERT is better than BioClinicalBERT. This is because of the different pre-trained data of the BERT model: the BlueBERT model learned from clinical notes in the MIMIC-III [36] database and the biomedical domains abstracts from PubMed [37], which is larger and has more comprehensive training datasets than BioClinicalBERT. This is why the BlueBERT model can achieve a better performance. Notably, our approach successfully addresses the challenge of mixed-language data, thus leading to greater consistency in the training data and enhancing the effectiveness of the model's learning process. We were also able to successfully integrate numerical features into the language model, thus enabling it to acquire a more comprehensive understanding of clinical knowledge. As a result, our model achieved the highest AUROC and AUPRC scores among the compared methods and demonstrated the most optimized performance. A high AUROC signifies that the model is proficient at distinguishing between positive and negative cases. This has real-world utility, as it implies that the model can make accurate predictions or classifications, leading to more informed decision-making. In healthcare, a high AUROC for a disease diagnostic model implies strong potential for early disease detection, improving patient outcomes. Moreover, high AUROC values indicate a balanced trade-off between sensitivity and specificity, which means the model minimizes both false positives and false negatives and thus serves as a powerful indicator of the model's overall effectiveness.

4. Discussion

From our literature review, our study is the first to predict emergency patient disposition and the possibility of unscheduled return after discharge at the same time. This provides better information than

Table 6
The performance results of the compared methods.

Models	Disposition						Unscheduled Return Visits					
	AUROC	AUPRC	Sensitivity	Specificity	PPV	NPV	AUROC	AUPRC	Sensitivity	Specificity	PPV	NPV
CNB	0.8159 (±0.0043)	0.6284 (±0.0078)	0.3464 (±0.0067)	0.9639 (±0.0017)	0.7495 (±0.0111)	0.8254 (±0.0016)	0.5895 (±0.0134)	0.0516 (±0.0083)	0.0856 (±0.0191)	**0.9804** (±0.021)	**0.0909** (±0.0166)	0.9791 (±0.0004)
KNN	0.7721 (±0.0035)	0.5935 (±0.0071)	0.3857 (±0.0065)	0.9391 (±0.0031)	0.6641 (±0.0133)	0.8305 (±0.0017)	0.5763 (±0.0217)	0.0518 (±0.0068)	0.1929 (±0.0190)	0.9018 (±0.0057)	0.0432 (±0.0044)	0.9799 (±0.0005)
RF	0.8494 (±0.0031)	0.6743 (±0.0062)	0.3084 (±0.0072)	0.9778 (±0.0014)	0.8125 (±0.0100)	0.8193 (±0.0016)	0.6091 (±0.012)	0.0490 (±0.0104)	0.0876 (±0.0215)	0.9798 (±0.0010)	0.0906 (±0.0192)	0.9791 (±0.0012)
XGB	0.8521 (±0.0041)	0.6905 (±0.0078)	0.4280 (±0.0105)	0.9590 (±0.0011)	0.7651 (±0.0051)	0.8431 (±0.0024)	0.5923 (±0.0125)	0.0440 (±0.0039)	0.1178 (±0.0156)	0.9570 (±0.0038)	0.0592 (±0.0071)	0.9793 (±0.0003)
LGBM	0.8569 (±0.0041)	0.6970 (±0.0077)	0.4412 (±0.0081)	0.9582 (±0.0017)	0.7671 (±0.0071)	0.8461 (±0.0018)	0.6104 (±0.0162)	0.0516 (±0.0079)	0.1122 (±0.0190)	0.9642 (±0.0025)	0.0670 (±0.0104)	0.9793 (±0.0004)
BioClinicalBERT	0.8797 (±0.1868)	0.7345 (±0.1551)	0.6211 (±0.1309)	0.9201 (±0.1958)	0.7085 (±0.1502)	0.8862 (±0.1884)	0.6262 (±0.0212)	0.0498 (±0.0064)	0.1905 (±0.0251)	0.9181 (±0.0189)	0.0523 (±0.0090)	0.9802 (±0.0003)
BlueBERT	0.8798 (±0.0038)	0.7349 (±0.0082)	0.6238 (±0.0115)	0.9208 (±0.0025)	0.7106 (±0.0063)	0.8870 (±0.0030)	0.6334 (±0.0157)	0.0510 (±0.0056)	0.2172 (±0.0378)	0.9085 (±0.0192)	0.0522 (±0.0057)	0.9806 (±0.0006)
Our method	**0.9014** (±0.2155)	**0.7739** (±0.1855)	**0.8037** (±0.1630)	0.8608 (±0.2290)	0.6760 (±0.1783)	0.9184 (±0.2233)	**0.6475** (±0.0202)	**0.0539** (±0.0056)	**0.2435** (±0.0251)	**0.8991** (±0.0132)	**0.0532** (±0.0081)	**0.9811** (±0.0006)

prediction of emergency patient disposition alone. When it comes to deciding patient disposition, patient clinical conditions along with the compliance of the medical advice are taken into account. Knowing patient disposition and the possibility of unscheduled return can help clinician decision making. For these reasons, our prediction model offers 2-step clinical decision making for physicians.

The first step is to predict whether the patients will be hospitalized, which can improve patient flow and mitigate ED overcrowding. For instance, if hospitalization is necessary for this patient, which suggests that his/her severity is higher than those who can be discharged, timely intervention and examinations will be arranged. Our prediction results can therefore act as an alarm for physicians. These patients will receive timely medical care, such as fluid resuscitation for septic patients [38], or certain administrative procedures could be arranged at the beginning to avoid wasting time. In this way, the process of hospitalization can be initiated at the beginning along with adequate treatment and examination at ED.

The second clinical step is to predict whether the discharged patients will return within 72 h, which reminds physicians about the importance of communicating with patients, adequately educating them, and properly managing them before they are discharged. For example, if clinicians are aware of the high possibility of the patient's unscheduled return, they should re-evaluate the patient's condition by reconsidering the discharge decision. If there is no further change of the patient's disposition, appropriate education and communication can be given to reassure patients' compliance and prevent unnecessary revisits. This will reduce medical costs and unnecessary examinations on the 2nd visit, which indicates better medical quality and will result in higher patient satisfaction. Although machine learning methods have been applied in previous studies which have widely used structured data such as vital signs, demographic data and laboratory results, it is unfortunate that some of these data, such as laboratory results, were not available right after triage, thereby making these methods less effective than those that could be implemented right away to improve ED patient flow. In our study, all the features applied in the model are routinely collected at triage, which means this model is not only applicable in any ED setting but also more effective.

Our model can predict emergency patient disposition with better performance compared to previous studies (AUROC 0.90 vs. 0.853, 0.785, 0.844 respectively) [5,39–41]. In line with previous research, we found that combining structured data and free text results in better performance (AUROC 0.90 vs. 0.88, respectively). However, the outperformance of the combination model was evident only after the structured data was translated into a narrative description. With the original numerical data, the performance was not as good as the one with free text alone (AUROC 0.79 vs. 0.88). The difference between free text only and free text with structured data is not so obvious. Based on free text alone, the patient's self-reported symptoms can also successfully predict the patient's disposition (AUROC 0.88). In that case, our model could even be applied in a pre-hospital setting, which suggests a possible solution for patient flow before ED triage.

As for the 72-hour unscheduled return visit prediction, the performance of our model is not as good as those in previous studies (AUROC 0.647 vs. 0.76, 0.755, 0.74 respectively). In our dataset, the incidence of unscheduled revisit (2.2 %) is lower than other studies (Table 1). The problem here is imbalanced data, which is a challenge for building predictive models and is due to skewed class distributions. We have applied the Synthetic Minority Oversampling Technique (SMOTE). SMOTE first selects a minority class instance a at random and finds its k nearest minority class neighbors. A synthetic instance is then created by choosing one of the k nearest neighbors b at random and connecting a and b to form a line segment in the feature space. The synthetic instances are next generated as a convex combination of the two chosen instances a and b. The default setting of SMOTE oversampling can make the partition of the positive instance equal to the negative instance. In our experiment, we used the oversampling of clinical narratives, vital signs,

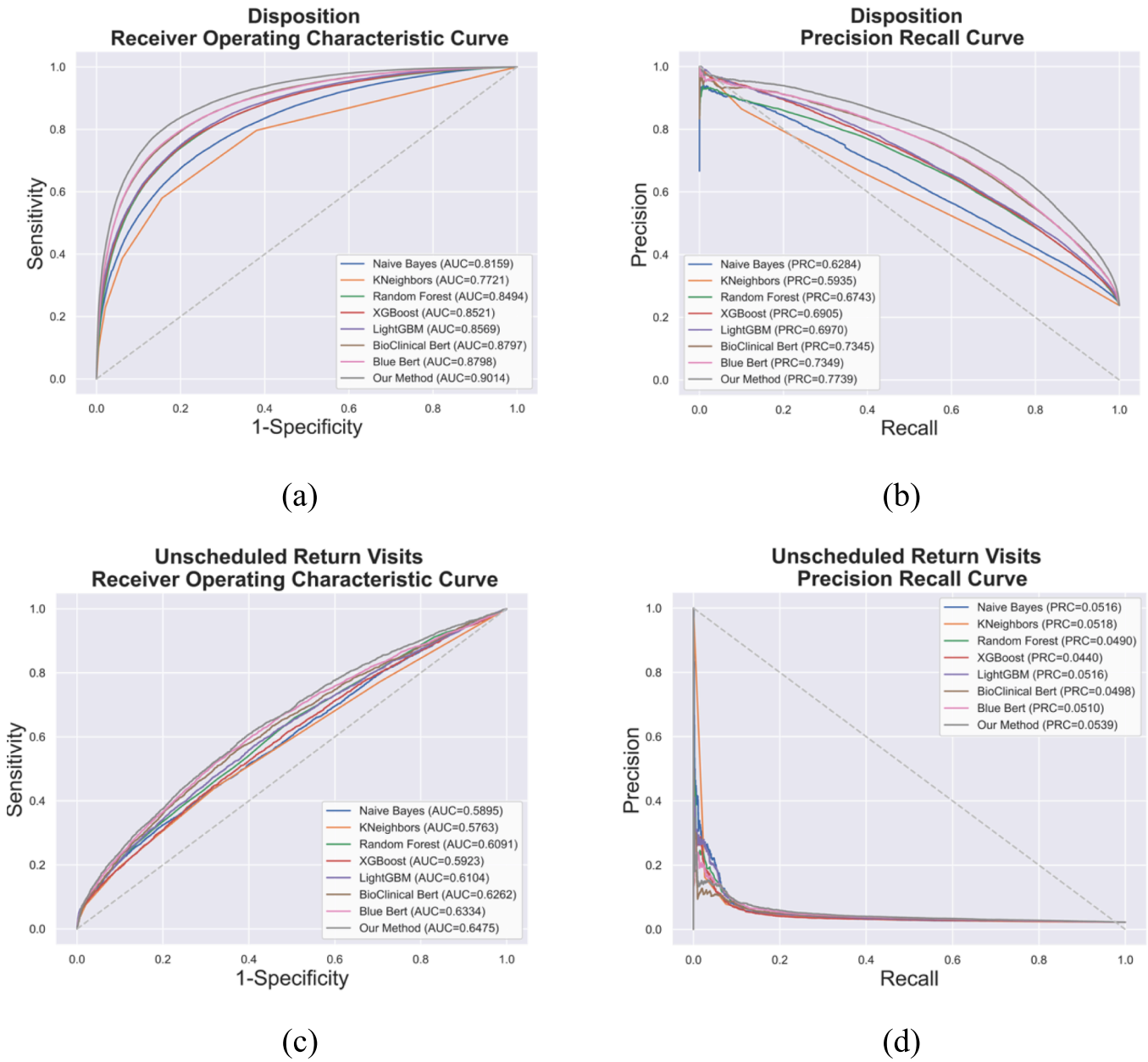

Fig. 3. The Receiver-operating-characteristics (ROC) curve and Precision-Recall (PR) curve of different methods for ED patient prediction under a 10-fold cross-validation. (a) ROC curve of patient disposition. (b) PR curve of patient disposition. (c) ROC curve of patient unscheduled return visits. (d) PR curve of patient unscheduled return visits.

and patient demographic data of minority classes so that the amount of data in both classes is balanced. However, even though we applied the SMOTE method to solve this problem, the performance was not as good as expected. In addition, our model's prediction of unscheduled revisit of all-cause emergency patients is subject to increasing difficulty with lower accuracy. Even though the specificity of our model is high (0.91), the sensitivity is poor. In other words, for the purposes of determining whether the patient will return to ER within 72 h, it's not very reliable. Models with better performances are those focusing on specific symptoms, such as abdominal pain or COVID-19 (Table 1). Therefore, if we modify our models to predict specific groups of similar symptoms, this may improve the performance; it will also be more feasible for clinical physicians to apply this knowledge in daily practice.

In this study, our model underwent a comprehensive evaluation by comparing it with five machine learning models and two pre-trained language models. The results of two clinical pre-trained language models, BioClinicalBERT and BlueBERT, showcased significantly superior capabilities compared to other machine learning-based methods. This underscores the effectiveness of pre-trained BERT models in extracting crucial information from text data. However, using only text data has its limitations, as it may not fully capture all the nuances present in clinical records. That is why we merged structured and free text data, converting numerical data into narrative form for analysis, which outperformed traditional numerical models (Table 5). This approach diverges from the usual practice of categorizing narrative data in medical research and highlights the efficacy of narrative descriptions over direct quantitative measurements in predicting outcomes. We categorized vital signs using clinical criteria, such as classifying consciousness level with the Glasgow Coma Scale (severe unconscious GCS 3–8, moderate GCS 9–14, clear GCS 15) and defining heart rate ranges for tachycardia (heart rate faster than 100 beats/min) and bradycardia (slower than 60 beats/min). This methodology aims at enhancing

clinical decision-making, particularly how physicians adjust treatments in response to significant changes in vital signs, like a notable drop in the GCS score. Our results showed that BERT, as a state-of-the-art language representation model, performs better on datasets consisting entirely of text data compared to datasets where text is combined with categorical variables. When the dataset consists entirely of text data, the BERT model can leverage its powerful natural language processing capabilities to convert words and semantic relationships into vector representations, extract contextual information and features, and make predictions. Therefore, datasets that combine text with categorical variables decrease the performance of the BERT model, since the categorical variables introduce noise and interference, thus making it difficult for BERT to capture the true features and relationships in the text. This means that transforming numerical data into a narrative corpus, rather than combining them, may result in the loss of some nuanced information. However, this approach represents a promising avenue for future research. Further, the effectiveness of pre-trained models can be constrained when applied to a mixed-language corpus, so it is essential to first translate mixed-language documents into a single language before proceeding with analysis. Finally, for the clinical decision-making system, it is critical to ensure that protocols are written in a single language to prevent any misinterpretation.

Despite our study's encouraging results, there are still a number of limitations. First, since our dataset comes from a single teaching hospital without external validation, overfitting should be taking into consideration. External validation with the dataset from another hospital in future studies would enhance the robustness of our findings. Second, hospitalization decisions primarily follow medical practice guidelines but are also shaped by hospital settings, resources, and doctor-patient relationships. These factors can vary by region, and for this reason suggests the need for models with broader, cross-national data for better generalizability. Finally, our data collection did not account for race and ethnicity, primarily because Taiwan is predominantly comprised of a single ethnic group, which is East Asian. This lack of diversity in our dataset limits the scope for incorporating data on race and ethnicity. Given the East Asian ethnicity in Taiwan, caution must be exercised when generalizing the results of our study to other ethnic or racial groups.

5. Conclusion

The conclusion of our study underscores the remarkable efficacy of the BlueBERT model in predicting patient hospitalization status and unplanned readmissions within Emergency Departments (EDs). Our research stands as contributions to the field of biomedical informatics, particularly in addressing the pressing issue of ED overcrowding. By utilizing a comprehensive dataset from Taipei Medical University-Shuang Ho Hospital and incorporating both structured and unstructured data, our approach has demonstrated not only high predictive accuracy but also practical applicability in clinical settings. The superior performance of BlueBERT, evidenced by its high AUROC scores, highlights the model's ability to effectively interpret and analyze complex medical data. This advancement is a testament to the potential of sophisticated natural language processing techniques in revolutionizing healthcare outcomes. Our findings indicate that the application of BlueBERT can significantly aid in reducing avoidable admissions and facilitating early preventive measures, ultimately contributing to more efficient and effective patient care in emergency settings.

This study, therefore, lays the groundwork for future research and development in the field of medical AI and offers a promising solution to mitigate the challenges of ED overcrowding. In our research, BlueBERT's success encourages further exploration and adoption of advanced AI models in healthcare, making it possible for clinicians to make smarter and data-driven decisions that can improve patient outcomes.

6. Ethics approval and consent to participate

This study was approved by Taipei Medical University Joint Institutional Review Board (TMU- JIRB). The number of certification is N202003035, which is as the attached document.

7. Consent for publication

This study did not contain any individual person's data in any form. Consent for publication was waived with permission from TMU-JIRB.

8. Availability of data and materials

The data that support the findings of this study are available from Shuang-Ho Hospital but restrictions apply to the availability of these data, which were used under license for the current study and so are not publicly available. Data are however available from the authors upon reasonable request and with permission of Shuang-Ho hospital.

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Author contributions

T.Y.H. and Y.C.C. participated in the original conception of the work. T.Y.H. and members of Shuang-Ho Hospital assisted with patient enrolment and data acquisition. T.Y.C. and Y.C.C. developed the architecture of our model, conducted the experiments, and analyzed the data. All authors drafted the original manuscript. All authors had access to and verified the raw data. All authors read and approved the final manuscript. All authors contributed feedback and approved the final draft, and no authors were precluded from accessing the data. All authors had final responsibility for the decision to submit for publication.

CRediT authorship contribution statement

Tzu-Ying Chen: Writing – review & editing, Writing – original draft, Visualization, Validation, Methodology, Investigation, Formal analysis. **Ting-Yun Huang:** Writing – review & editing, Writing – original draft, Validation, Formal analysis, Data curation, Conceptualization. **Yung-Chun Chang:** Writing – review & editing, Writing – original draft, Supervision, Software, Resources, Project administration, Methodology, Funding acquisition, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Statement of Significance

Problem: The escalation of overcrowding in Emergency Departments (EDs) undermines the delivery of critical healthcare services,

necessitating effective mitigation strategies to manage patient flow and outcomes efficiently.

What is already known: Previous approaches to predicting patient disposition and unscheduled return visits in EDs have leveraged various predictive models, but many lack the integration of deep learning and natural language processing (NLP) techniques, specifically tailored to the biomedical domain, which can significantly enhance the prediction accuracy by understanding the nuanced and complex nature of clinical narratives.

What this paper adds: This study introduces a groundbreaking application of BlueBERT, a biomedical domain-specific pre-trained language model, for predicting emergency department patient disposition and the likelihood of unscheduled return visits. By utilizing a large dataset of ∼ 250,000 emergency department visit records from Taipei Medical University-Shuang Ho Hospital and incorporating both structured and unstructured data, our model significantly outperforms traditional predictive models. The research demonstrates the superior efficacy of BlueBERT, with high AUROC scores indicating its remarkable prediction capabilities in a clinical setting. Our findings emphasize the model's potential in reducing ED overcrowding by facilitating more accurate patient disposition decisions and identifying patients at risk of unplanned readmissions. This advancement not only enhances patient care but also optimizes resource allocation in emergency settings. This paper lays the foundation for future advancements in medical AI, advocating for the adoption of sophisticated NLP techniques in healthcare to improve patient outcomes and address the critical issue of ED overcrowding.

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